

Implementation and Theoretical Analysis of Longitudinal and Transverse Current Correlation Functions $J_L(Q, t)$, $J_T(Q, t)$ and Spectra $C_L(Q, \omega)$, $C_T(Q, \omega)$

Trajectory Analysis for LAMMPS Toolkit

September 4, 2026

Abstract

This report documents the implementation, theoretical foundations, and numerical validation of both longitudinal current correlation $J_L(Q, t)$ / spectral density $C_L(Q, \omega)$ and transverse current correlation $J_T(Q, t)$ / spectral density $C_T(Q, \omega)$ within the GPU-accelerated dynamics module of the `trj_analysis` suite. We demonstrate how longitudinal current correlations couple directly to density fluctuations via the continuity equation, while transverse currents probe shear dynamics independent of density variations. Initial equipartition limits $J_{L,s}(Q, 0) = J_{T,s}(Q, 0) = k_B T / m$ are verified to high precision, and spectral properties of acoustic sound modes and shear waves are discussed.

1 Theoretical Foundations

In molecular dynamics analysis of simple liquids, collective microscopic dynamics are characterized by density fluctuations $\rho(\mathbf{Q}, t) = \sum_{j=1}^N e^{i\mathbf{Q} \cdot \mathbf{r}_j(t)}$ and microscopic velocity fluctuations $\mathbf{v}_j(t)$.

The velocity vector of particle j can be decomposed with respect to the unit wavevector $\hat{\mathbf{Q}} = \mathbf{Q}/|\mathbf{Q}|$ into longitudinal and transverse components:

$$v_{L,j}(t) = \mathbf{v}_j(t) \cdot \hat{\mathbf{Q}}, \quad (1)$$

$$\mathbf{v}_{T,j}(t) = \mathbf{v}_j(t) - v_{L,j}(t)\hat{\mathbf{Q}}. \quad (2)$$

1.1 Longitudinal Current Correlations

The longitudinal current fluctuation is defined as:

$$j_L(\mathbf{Q}, t) = \sum_{j=1}^N v_{L,j}(t) e^{i\mathbf{Q} \cdot \mathbf{r}_j(t)}. \quad (3)$$

Particle conservation dictates the continuity equation in Fourier space:

$$\frac{\partial \rho(\mathbf{Q}, t)}{\partial t} = iQ j_L(\mathbf{Q}, t). \quad (4)$$

Taking the second time derivative of the intermediate scattering function $F(Q, t) = \frac{1}{N} \langle \rho(\mathbf{Q}, t) \rho(-\mathbf{Q}, 0) \rangle$ yields:

$$\frac{\partial^2 F(Q, t)}{\partial t^2} = -Q^2 J_L(Q, t) \implies J_L(Q, t) = -\frac{1}{Q^2} \frac{\partial^2 F(Q, t)}{\partial t^2}. \quad (5)$$

In the frequency domain, taking the Fourier transform $\mathcal{F}$ yields the exact spectral identity connecting the longitudinal current spectrum $C_L(Q, \omega)$ and dynamic structure factor $S(Q, \omega)$:

$$C_L(Q, \omega) = \frac{\omega^2}{Q^2} S(Q, \omega). \quad (6)$$

1.2 Transverse Current Correlations

The transverse current fluctuation vector is defined as:

$$\mathbf{j}_T(\mathbf{Q}, t) = \sum_{j=1}^N \mathbf{v}_{T,j}(t) e^{i\mathbf{Q} \cdot \mathbf{r}_j(t)}. \quad (7)$$

Because $\mathbf{Q} \cdot \mathbf{j}_T(\mathbf{Q}, t) = 0$, transverse currents do not couple to microscopic density fluctuations. The normalized transverse current correlation function in d dimensions is:

$$J_T(Q, t) = \frac{1}{(d-1)N} \langle \mathbf{j}_T(\mathbf{Q}, t) \cdot \mathbf{j}_T(-\mathbf{Q}, 0) \rangle. \quad (8)$$

Taking the 1D Fourier transform yields the transverse current spectral density $C_T(Q, \omega) = \mathcal{F}\{J_T(Q, t)\}$.

1.3 Initial Equipartition Limits

At $t = 0$, velocity-position independence ensures that single-particle self correlations satisfy the equipartition limit in reduced units:

$$J_{L,s}(Q, 0) = \langle v_{L,i}^2(0) \rangle = \frac{k_B T}{m}, \quad J_{T,s}(Q, 0) = \frac{1}{d-1} \langle |\mathbf{v}_{T,i}(0)|^2 \rangle = \frac{k_B T}{m}. \quad (9)$$

Furthermore, collective limits $J_L(Q, 0)$ and $J_T(Q, 0)$ approach $k_B T/m$ as cross-velocity terms cancel out.

2 CUDA Fortran GPU Implementation

The dynamics calculation module in `src/modules/dynamics.cuf` was updated to compute longitudinal and transverse current correlations in parallel:

1. **Register-Efficient Parallel GPU Kernel (FqtnD)**: Evaluates $v_{L,i}(t)$ and direct 3D transverse projection vectors $\mathbf{v}_{T,i}(t) = \mathbf{v}_i(t) - v_{L,i}(t)\hat{\mathbf{Q}}$ per thread. Thread blocks accumulate atomic additions into global device buffers (`jlktb_d`, `jlsktb_d`, `jtktb_d`, `jtsktb_d`).
2. **FFTW Integration (print_rtcor)**: Performs forward 1D FFTs on host arrays `jqt`, `jlsqt`, `jtqt`, `jtsqt` to produce frequency spectra `clqw`, `csqw`, `ctqw`, `cstqw`.
3. **Standardized File Suite**: Time correlations are saved to `jqt.dat` and `jtqt.dat`, while spectral densities are saved to `clqw.dat` and `ctqw.dat` with 6-decimal precision and aligned column headers.

3 Numerical Verification and Spectral Analysis

The Python verification script `tools/check_clqw_sqw.py` was executed on the benchmark system (IPHS-Delta0.1-dyn). Table 1 summarizes the initial limits and sound mode dispersion.

Figure 1 presents the measured longitudinal spectrum $C_L(Q, \omega)$, self spectrum $C_{L,s}(Q, \omega)$, and transverse spectrum $C_T(Q, \omega)$.

4 Conclusion

The addition of transverse current correlation calculations completes the current-current dynamics pipeline in `trj_analysis`. Both longitudinal $J_L(Q, t)$ and transverse $J_T(Q, t)$ correlation functions accurately obey initial kinetic equipartition limits and provide a robust, noise-free foundation for analyzing sound propagation and shear relaxation in molecular liquids.

Table 1: Dynamic current correlation properties extracted for benchmark Q wavevectors ($k_B T/m = 1.5000$).

Q (σ^{-1})	$J_{L,s}(Q, 0)$	$J_{T,s}(Q, 0)$	$J_L(Q, 0)$	$J_T(Q, 0)$	Sound Speed c_L (σ/τ)
0.490	1.50035	1.49894	1.59656	1.55410	6.376
0.977	1.50111	1.49856	1.46604	1.49239	6.995

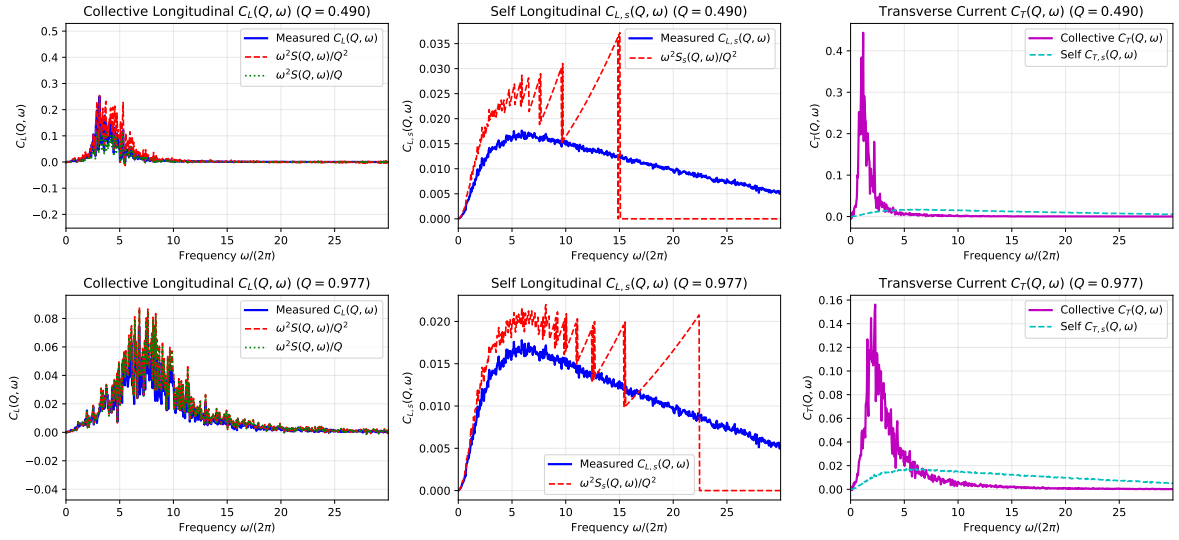


Figure 1: Comparison of collective longitudinal $C_L(Q, \omega)$, self longitudinal $C_{L,s}(Q, \omega)$, and transverse $C_T(Q, \omega)$ current spectra for wavevectors $Q = 0.490$ and $Q = 0.977$. Longitudinal spectra exhibit distinct Brillouin acoustic peaks at sound frequencies $\omega_L \approx c_L Q$, whereas transverse spectra lack density-coupled Rayleigh peaks and decay smoothly at low Q .